

TGFB1 promoter-reporter architectures

5' -> 3' display, genomic 41358961 -> 41287203 | strand - | assembly GRCh38 | annotation Ensembl live GRCh38 | upstream 5000 bp | downstream 1800 bp

Canonical reporter comparison: 1 TSS class(es), 3 transcript(s), 9 explicit architecture row(s); normalized evidence remains observational/predictive context.

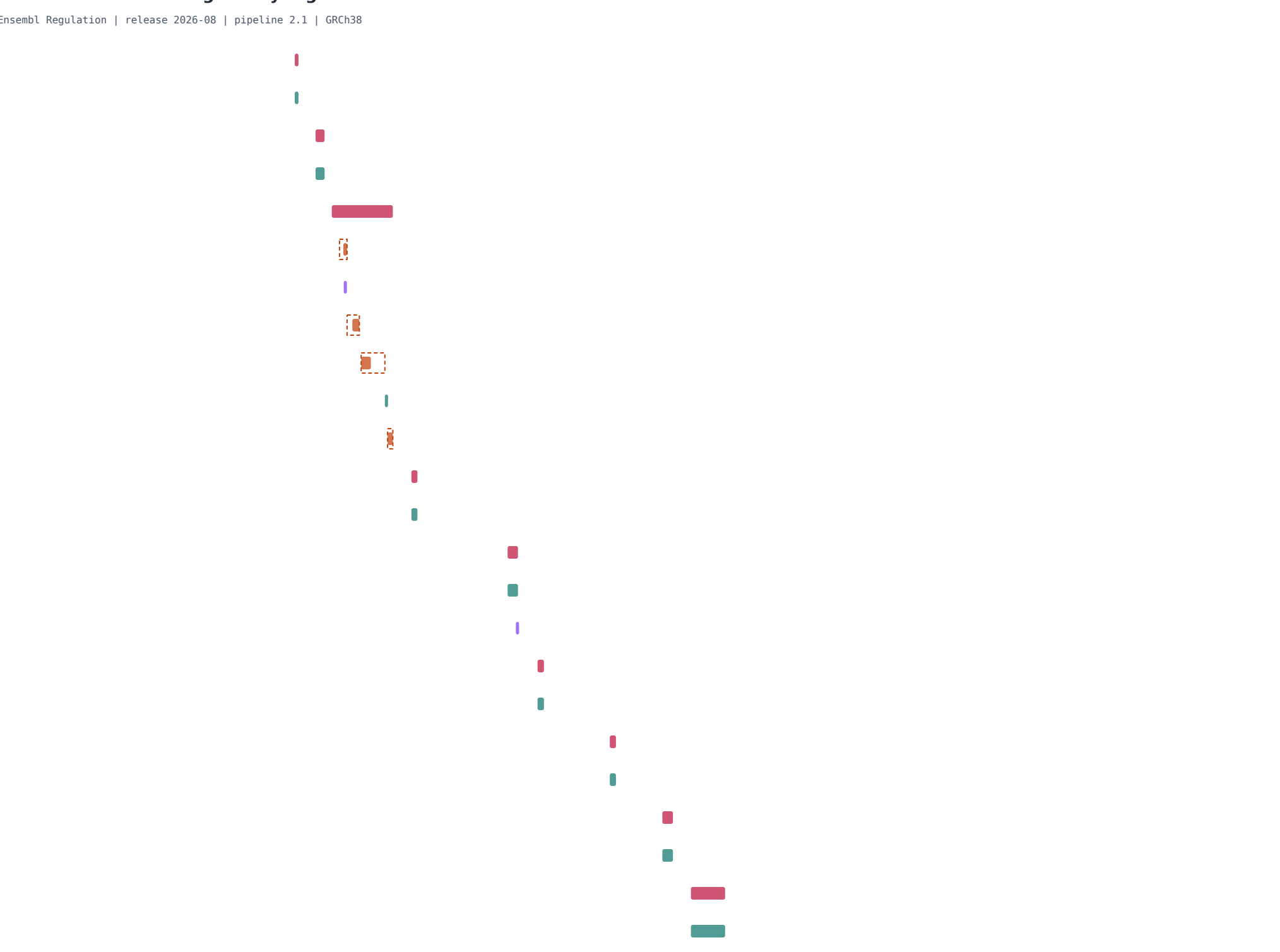


Proposed reporter architectures



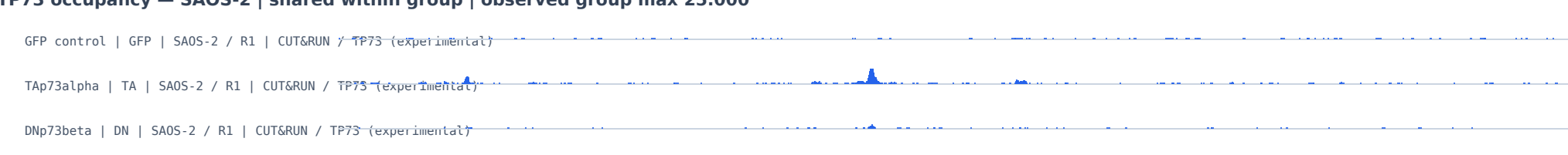
Ensembl-annotated regulatory regions

Ensembl Regulation | release 2026-08 | pipeline 2.1 | GRCh38

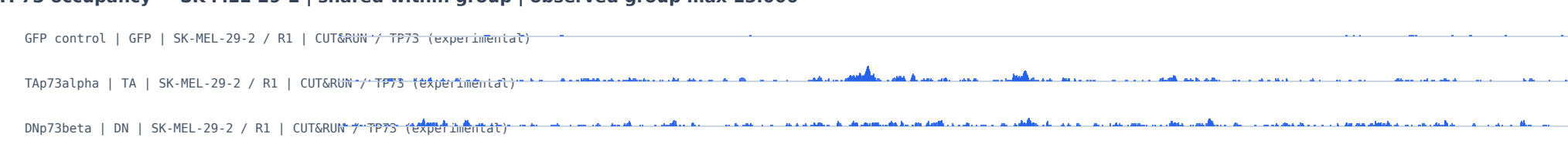


Projected occupancy grouped by declared biological context

TP73 occupancy — SAOS-2 | shared within group | observed group max 25.000



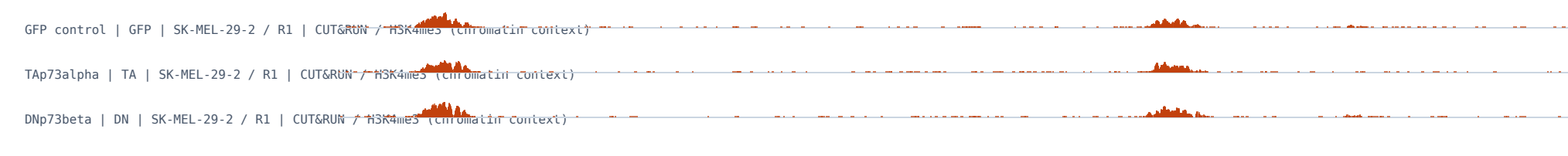
TP73 occupancy — SK-MEL-29-2 | shared within group | observed group max 13.000



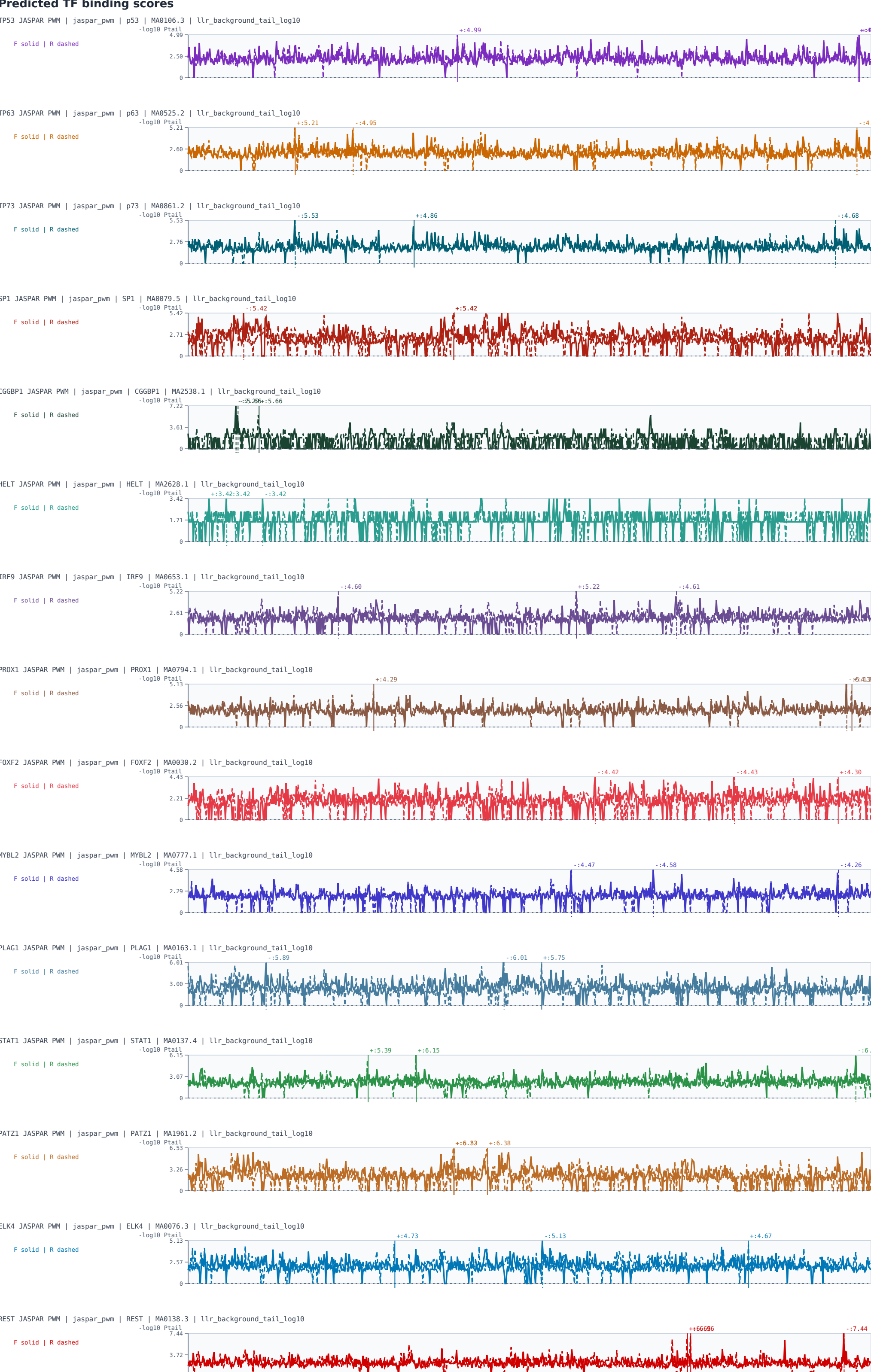
H3K4me3 chromatin context — SAOS-2 | shared within group | observed group max 31.000



H3K4me3 chromatin context — SK-MEL-29-2 | shared within group | observed group max 32.000



Predicted TF binding scores



TSS-local Ensembl-feature promoterome similarity

Transcript-oriented -500/+200 bp; features clipped to the displayed stretch; BLASTN >40 bp, >80% identity, E=1e-05; left-locus and same-gene targets excluded.

Blue frequency strips count distinct other genes by query position. Numbered blocks give target-promoter order; red outlines mark order/orientation breaks.

TGFB1_tss_stretch_1



Similarity is structural evidence; Ensembl classes, predicted TFBS and CUT&RUN enrichment do not prove reporter activity or sufficiency.

Reporter interpretation boundaries

Transcript annotation alone does not establish which TSS is used in the assayed cells.

CUT&RUN overlap is occupancy/enrichment evidence, not proof of TSS usage, promoter activity, direct regulation, biochemical affinity, or luciferase output.

Reporter geometry does not verify the identity of a physical laboratory construct.

Warnings / interpretation checks

Evidence provenance

ensembl_regulation_annotation_index: ensembl_regulation_2026_08_grch38
ensembl_regulation_annotation_intervals: ensembl_regulation_2026_08_grch38
isoform_panel: TGFB1_panel
occupancy_layout: tp73_saos2,tp73_skmel29-2,h3k4me3_saos2,h3k4me3_skmel29-2
predicted_tf_score_jasper_pwm: local_jasper_registry
projected_occupancy_track: E-HTAB-15789:h3k4me3:saos2:DN:R1
projected_occupancy_track: E-HTAB-15789:h3k4me3:saos2:GFP:R1
projected_occupancy_track: E-HTAB-15789:h3k4me3:saos2:TA:R1
projected_occupancy_track: E-HTAB-15789:h3k4me3:skmel29-2:DN:R1
projected_occupancy_track: E-HTAB-15789:h3k4me3:skmel29-2:GFP:R1